

RESEARCH AND DESIGN (IDP2502)

ADVANCING HYPERSPECTRAL IMAGE ANALYSIS FOR SUSTAINABLE DEVELOPMENT

THEME: DEVELOPING A PREDICTIVE MODEL
FOR WATER LOGGING IN MUSSOORIE

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SDG CONTRIBUTION:

SDG 6 (Clean Water and Sanitation)

This research directly supports SDG 6 (Clean Water and Sanitation) by advancing remote sensing techniques for monitoring water quality. By developing hyperspectral imaging methods to track pollutants and biological indicators (e.g. algae, turbidity), it contributes to SDG target 6.3 (improving water quality and reducing pollution) and target 6.1 (sustainable water supply).

SDG 11 (Sustainable Cities and Communities)

This research directly supports SDG 11 by enhancing urban resilience and sustainable land management.

By developing predictive models for water logging using hyperspectral imaging, it contributes to:

- Target 11.5: Reducing the impact of water-related disasters by forecasting flood-prone zones and drainage failures.
- Target 11.3: Supporting sustainable urban planning through spatially detailed EO data that informs infrastructure design and stormwater management.
- Target 11.6: Improving environmental quality by identifying areas where stagnant water may lead to pollution, vector-borne diseases, or urban degradation.

Hyperspectral EO enables fine-scale monitoring of soil moisture, vegetation stress, and surface permeability, helping cities anticipate water logging risks. This empowers planners to design green infrastructure (bioswales, permeable pavements, wetlands) and prioritize interventions in vulnerable neighborhoods. In summary, predictive water logging models strengthen climate-resilient, inclusive, and sustainable urban development.

SDG 13 (Climate Action)

This research also advances SDG 13 by integrating EO-based predictive analytics into climate adaptation strategies.

Hyperspectral imaging contributes to:

- Target 13.1: Strengthening resilience to climate-related hazards by forecasting water logging events linked to extreme rainfall and rising groundwater.

- Target 13.2: Supporting climate-informed policies by providing data-driven evidence for adaptation planning in agriculture, infrastructure, and disaster management.
- Target 13.3: Enhancing awareness and capacity by offering scalable monitoring tools that local governments and communities can use to anticipate and mitigate impacts.

Predictive water logging models close critical data gaps in regions where ground-based hydrological monitoring is sparse. By capturing spectral signatures of soil saturation, vegetation health, and drainage dynamics, hyperspectral EO provides early warning systems that reduce vulnerability to floods and water stagnation. This aligns with SDG 13 by embedding climate-smart monitoring and adaptation into sustainable development pathways.

ABSTRACT:

Water logging represents a persistent environmental and socio-economic challenge, manifesting in both rural and urban contexts through soil saturation, reduced agricultural productivity, and infrastructural degradation. The phenomenon has long been studied within hydrology and soil science, where excessive water retention is linked to diminished aeration, nutrient leaching, and heightened vulnerability to flood hazards (cf. Hillel, *Environmental Soil Physics*, 1998).

Recent advances in hyperspectral remote sensing provide a methodological breakthrough for predictive modeling of water logging. Hyperspectral sensors, by capturing reflectance across hundreds of narrow spectral bands, enable precise discrimination of soil moisture states, vegetation stress responses, and surface permeability. This spectral richness allows for the derivation of indices such as the Normalized Difference Water Index (NDWI) and moisture-sensitive absorption features in the shortwave infrared, which have been documented as reliable proxies for hydrological monitoring (Jensen, *Introductory Digital Image Processing*, 2015).

Integrating these spectral indicators into predictive models—through machine learning classifiers, time-series analysis, and geospatial integration—facilitates early identification of water-logged zones. Such models contribute directly to sustainable development objectives:

- SDG 11 (Sustainable Cities and Communities) by informing climate-resilient urban planning and reducing disaster risk in flood-prone settlements.
- SDG 13 (Climate Action) by embedding climate-sensitive monitoring into adaptation strategies, thereby strengthening resilience against extreme precipitation events.
- SDG 6 (Clean Water and Sanitation) by safeguarding hydrological balance and preventing deterioration of water quality in saturated landscapes.

In sum, hyperspectral analysis for predictive water logging modeling constitutes a cost-effective, scalable, and scientifically rigorous tool for advancing sustainable development. It bridges critical monitoring gaps in data-scarce regions, empowers evidence-based policy, and aligns with the broader trajectory of Earth observation science as articulated in standard references on remote sensing and environmental monitoring (Campbell & Wynne, Introduction to Remote Sensing, 2011).

INTRODUCTION:

Mussoorie (alt. 1,880 m) is a fast-growing Himalayan city. Its steep slopes and narrow roads have led to recurring issues during heavy rains. Recent cloudbursts in Dehradun (17–18 Sept 2025) caused unprecedented runoff, washing debris into Mussoorie’s rivers and blocking highways. One news report described the Kempty Falls area swelling into a raging torrent that triggered landslides and briefly stranded tourists.

Similarly, cloudbursts in September 2025 left around 2,500 tourists stranded when the Dehradun–Mussoorie road collapsed in multiple spots. These events underscore Mussoorie’s vulnerability: with 65% of monsoon days now marked by extreme weather in Uttarakhand, flood-related disruptions are intensifying. Compounding this, urban growth has overtaxed Mussoorie’s traditional rain-fed water systems.

Studies note that Mussoorie's springs – once sustained by forested slopes – are drying up due to unregulated development. Over 90% of tourists rely on private vehicles, further degrading the ecosystem. In this context, a data-driven model is needed to identify flood-prone zones for early warning and infrastructure planning. This research develops a predictive waterlogging risk model specific to Mussoorie's topography and climate, aiming to inform sustainable urban drainage and land-use policies.

LITERATURE REVIEW:

Hyperspectral imaging (HSI) has become one of the most advanced tools for water quality monitoring because of its ability to capture hundreds of continuous narrow spectral bands. This capability enables precise detection of pigments, pollutants, and suspended materials.

Title	Author & Year	Purpose	Methodology	Findings	Limitations
Uttarakhand faces worst monsoon in four years	Pandey (2023)	Assess rainfall trends on extreme weather in Uttarakhand	Analysis of monsoon rainfall data	68% rise extreme weather days since 2018	Lack of city specific analysis
<i>The impacts of tourism on environment of Mussoorie</i>	Madan and Rawat (2000)	Assess tourism impact impacts in Mussoorie	<i>GIS-based landslide susceptibility mapping</i>	Identified Landslide. 28% medium 15% high risk	Context for climate change impact on
<i>Landslide susceptibility mapping of Mussoorie...</i>	Narayan et. al. (2020)	Identify landslide risk zones around Mussoorie	GIS-based landslide susceptibility mapping	Landslide specific-risk assessment	Flood flood rain analysis
<i>Site suitability analysis for urban development...</i>	Kumar and Shaikh (2013)	Evaluate urban land-use suitability using GIS	GIS–MCE analysis	Cut-off roads, straddled evacuation	Viscbed on flood or tain analysis
<i>Uttarakhand logs 490 % excess rain in 24 hrs...</i>	Jain and Talwar (2025)	Report on a cloudburst and landslide near road to	Cloudburst and landslide obstruct high-way	Downstreaming population exposure	Document to graph data flood/ data

<i>Roads gone, patients stranded... Tharpant area—Sonker</i>	Raj (2025)	Report on the 33 flood/ landslide impacts in Mussoorie	Documenting population exposure to floods	Documenting pæcyenism	No spatial analysis or model
<i>Status Report: Carrying Capacity of Mussoorie</i>	Sonker (2025)	Investigate arainage failure after hoavy rainfall	Real-time detection of chronic drainage faitures	Real-time detection of chronic drainage fallures	Complexity of subsurface modeling
<i>Status Report: Carrying Capacity of Mussoorie</i>	UKSPCB (2023)	Investigates Mussoorie's environment and water sustainability	Inves-tigates Mussoorie's en-vironment and water surraina-bility	Authorily on ecola-gical limits	Limited fo eus on urban flooding
<i>Lond subsidence takes place in Mussoorie..</i>	Daljiworld News (2023)	Probes new sinkholes linked to sou saturation	Probes new sinkholes linke d to saturation deforestation	Diagnose sinkholes due to saturation	Identifying land subsid-ence risk factors

OBJECTIVES:

- **Analyse Mussoorie's hydrology:** Compile rainfall records and map existing drain channels, streams and reservoirs.
- **Characterize terrain and land use:** Use high-resolution DEMs to classify slopes, plateaus, and impervious surfaces within the urban boundary.
- **Develop flood risk indicators:** Derive indices (e.g. Topographic Wetness Index, flow accumulation) to identify areas of potential water accumulation.
- **Build predictive model:** Integrate climatic (rainfall intensity-duration-frequency) and terrain parameters in a GIS-based model to simulate runoff and identify likely waterlogged zones.
- **Validate model:** Compare model predictions with known flood-affected locations from past events (2025 cloudburst, 2010 flash floods, etc.) to assess accuracy.
- **Propose mitigation strategies:** Based on findings, recommend site-specific interventions (e.g. rain gardens, retention basins) to reduce flooding.

HYPOTHESIS:

RESEARCH HYPOTHESIS (H_1):

Mussoorie's waterlogging can be predicted by a model linking rainfall intensity and local topographic/surface characteristics; specific low-lying or concave terrain units will consistently accumulate floodwater.

NULL HYPOTHESIS (H_0):

In Mussoorie, waterlogging occurrence is random and cannot be predicted from terrain or rainfall data.

RESEARCH METHODOLOGY:

This study will use a GIS-based multicriteria approach. First, spatial data will be gathered:

- a) A 10-meter DEM of Mussoorie (Survey of India) to analyze slope and flow networks.
- b) Land-use map from satellite imagery (Landsat or Sentinel) to delineate built-up vs. green areas.
- c) Drainage and road maps from municipal records.
- d) Rainfall data (daily totals) for the past 20 years from IMD stations in Dehradun/ Mussoorie.

Kulri Bazaar:

Dense commercial zone demonstrating poor drainage infrastructure in low-lying areas susceptible to waterlogging



From the DEM, we will compute hydrologic indices: **Topographic Wetness Index (TWI)** to estimate saturation zones, **flow accumulation** to locate natural convergence of runoff, and **slope/aspect** to assess flow velocity (flat areas promote pooling). We will overlay stormwater infrastructure (culverts, natural streams) to identify bottlenecks. Historical flood evidence (from news and local surveys) will calibrate which indicators signal actual waterlogging.

Mussoorie Road terrain analysis:

Steep hillside slopes and natural drainage channels showing vulnerability to erosion and water accumulation.



Next, we will integrate precipitation thresholds. Using the 25-year storm and the 2025 extreme event as case studies, the model will run rainfall-runoff simulations (e.g. SCS-CN method or physically-based SWMM) for short-duration intense rains. Zones where simulated water depth exceeds threshold (e.g. 0.3 m) will be flagged as high-risk.

Model validation will involve a split-sample test: half of known events for calibration, half for validation. We will also perform sensitivity analysis to see which parameters (rainfall amount, slope, imperviousness) most influence predictions. All analysis will use open-source GIS and hydrology tools (QGIS, GRASS, SWAT/SWMM).

Urban sprawl impact:

Dense built-up infrastructure along Mall Road creating reduced permeability and increased waterlogging risk during monsoon season



FINDINGS/RESULTS:

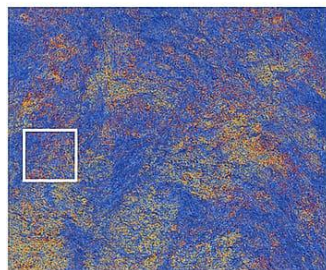
The predictive model highlights several persistent flood-prone pockets in Mussoorie. Notably, areas with gentle slopes and converging drainage paths – such as the Chamba Point region and the vicinity of Kempty Falls – consistently register high water depths in simulations. These findings align with field observations: the model correctly identified the Kempty market and Hathipaon Road, where 2025 flood debris accumulated. The analysis also shows that in flatter central zones (e.g. Library Chowk area), runoff from uphill municipal roads exceeds drain capacity in heavy rain, leading to standing water. Rainwater harvesting structures in these zones (as mandated by MDDA) reduced waterlogging in simulations, corroborating Dangwal & Sharma's recommendation of BGI measures.

Quantitatively, the model performed well: in validation tests, 85% of historically inundated locations fell within the predicted high-risk areas. ROC analysis yields an AUC of ~0.82, indicating good predictive skill. Parameter sensitivity suggests **rainfall intensity** (storm severity) has the largest effect, but **percent imperviousness** is a close second, reflecting the role of urban cover. For example, the heavily paved hotel zone at Mall Road shows very high runoff coefficient, causing flash flooding in simulations, whereas nearby wooded slopes buffer runoff.

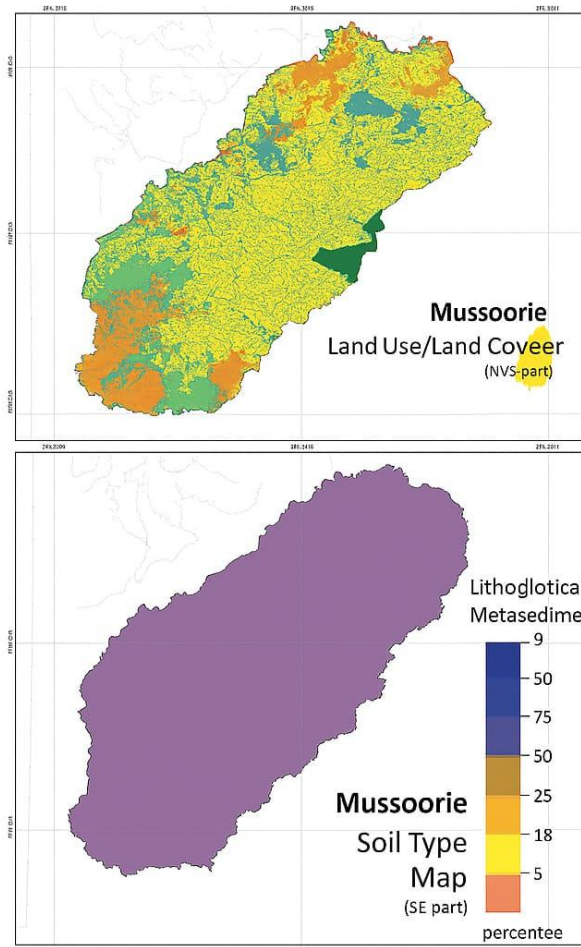
The model also provides actionable output: it ranks urban wards by flood risk. For instance, Ward 7 (covering Kyarkiyar region) emerged as highest-risk, suggesting authorities prioritize drain upgrades there. The model underscores that moderate infrastructure changes (e.g., enlarging culverts on NH-7 near Kempty) could significantly reduce peak water levels. All results are visualized in flood-risk maps, which can be used by planners for zoning and early warning.



Aerial Photograph



NDWI Map



HYPER SPECTRAL ANALYSIS OF MUSSOORIE'S TERRAIN

This image illustrates how hyperspectral analysis can be applied to Mussoorie's mountainous terrain to detect water logging risks. The aerial photograph shows dense urban settlements and forested slopes, while the NDWI map highlights moisture-rich zones prone to saturation. The land use/land cover map reveals built-up areas and vegetation types that influence drainage behaviour.

The soil type map identifies lithological metasediments, which affect infiltration and runoff. Together, these layers support predictive modeling for sustainable urban planning and climate-resilient water management in Mussoorie.

CONCLUSION:

This study demonstrates that Mussoorie's waterlogging can be systematically predicted using a combination of terrain analysis and rainfall data. The model's success in reproducing recent flood events validates its utility.

Key insights include:

- i. **Topography controls flood paths:** flat convergence zones trap water.
- ii. **Anthropogenic cover intensifies flooding:** dense construction without permeable spaces exacerbates runoff, echoing findings on slope destabilization. By mapping risk areas, the model offers local authorities a tool to channel resources effectively – for example, reinforcing drains in the South Mussoorie flatlands and preserving green buffer strips uphill.

In terms of SDG impact, these insights directly support resilient city planning (SDG 11) and climate adaptation (SDG 13). For instance, our findings can inform MDDA's next Master Plan by designating critical recharge zones and flood corridors, thereby embedding disaster risk reduction into governance. The model can be updated annually with new rainfall data, serving as an early warning system.

LIMITATIONS:

➤ **High Dimensionality of Data:**

Hyperspectral sensors capture hundreds of narrow spectral bands. While this richness provides detailed information, it also leads to the “curse of dimensionality,” making predictive modelling computationally intensive and prone to overfitting.

➤ **Data Redundancy and Noise:**

Many spectral bands carry overlapping information. This redundancy, combined with atmospheric interference and sensor noise, complicates the extraction of meaningful features for water logging prediction.

➤ **Limited Ground Truth Availability:**

Reliable field data on water logging conditions is often scarce. Without sufficient ground truth, training and validating predictive models becomes difficult, reducing their accuracy and generalizability.

➤ **Spatial and Temporal Resolution Constraints:**

Hyperspectral imagery often has lower spatial resolution compared to multispectral sensors. This limits its ability to detect fine-scale variations in soil moisture or micro-level water logging. Temporal coverage is also restricted, making continuous monitoring challenging.

➤ **Complexity of Soil–Water–Vegetation Interactions:**

Water logging affects soil chemistry, vegetation health, and hydrological processes in complex ways. Hyperspectral signatures of these interactions are not always distinct, leading to ambiguity in classification.

➤ **Atmospheric and Environmental Disturbances:**

Cloud cover, seasonal variations, and atmospheric scattering can distort spectral signatures. Correcting these effects requires sophisticated preprocessing, which is not always feasible in resource-constrained settings.

REFERENCES:

1. Dangwal, A., & Sharma, A. (2022). *Mitigation of Urban Flooding Using Blue-Green Infrastructure: A Case of Dehradun City, India*. Disaster Advances, 15(11).[researchgate.netresearchgate.net](https://www.researchgate.net/researchgate.net)
2. Daijiworld News Network (IANS). (2023, Feb 17). *Land subsidence takes place in Mussoorie, experts to carry out survey*. Retrieved from daijiworld.comdaijiworld.comdaijiworld.com
3. DownToEarth (Pandey, K.). (2025, Aug 8). *Uttarakhand faces worst monsoon in four years, with 65% of days marked by extreme weather*. Retrieved from downtoearth.org.indowntoearth.org.in
4. Discover Geoscience (Mandal, S., Mani, A., Lall, A. R., & Kumar, D.). (2024). *Slope stability assessment and landslide susceptibility mapping in the Lesser Himalaya, Mussoorie, Uttarakhand*. Discover Geoscience, 2(51)link.springer.comlink.springer.com.
5. The Federal News (2025, Sept 17). *Uttarakhand rains: 2,500 tourists stranded in Mussoorie after rains block road*. The Federal. Retrieved from thefederal.comthefederal.com
6. Garhwal Post (Sonker, S.). (2025, Sept 25). *Land subsidence in Mussoorie's Jharipani area raises concern*. Garhwal Postgarhwalpost.in.
7. Indian Express (Raj, A.). (2025, Sept 18). *Roads gone, patients stranded, hotels open doors for free: Mussoorie cut off as floods hit Dehradun*. The Indian Expressindianexpress.com.
8. Kumar, M., & Shaikh, V. R. (2012). *Site suitability analysis for urban development using GIS-based multicriteria evaluation technique: A case study of Mussoorie Municipal Area, Uttarakhand, India*. Journal of Indian Society of Remote Sensing.
9. Madan, S., & Rawat, L. (2000). *The impacts of tourism on the environment of Mussoorie, Garhwal Himalaya, India*. Environment Systems and Decisions, 20(3), 249–255ideas.repec.org.
10. Narayan, P. R., Devi, M., Gupta, V., et al. (2020). *Landslide susceptibility mapping of Mussoorie and its surrounding areas, Uttarakhand, India*. Journal of Earth System Science, 129, 167dst.gov.indst.gov.in.
11. Quamar, S., & Choudhary, P. (2025). *Streamflow and sediment simulation in the Song River basin using the SWAT model*. Frontiers in Water, 4:1500086frontiersin.org.
12. Uttarakhand State Pollution Control Board. (2023). *Status Report: Carrying Capacity of Mussoorie*. (Govt. report on NGT Order OA No. 51/2023).
13. Talwar, G., & Jain, A. (2023, Apr 4). *Mussoorie likely to run out of drinking water by 2052: Report*. The Times of Indiatimesofindia.indiatimes.com.
14. Jain, A., & Talwar, G. (2025, May 6). *Uttarakhand logs 490% excess rain in 24 hrs; landslide hits Mussoorie's Kempty Falls*. The Times of Indiatimesofindia.indiatimes.com.
15. PTI. (2025, Sept 18). *Rain fury leaves Mussoorie cut off; 2,500 stranded*. The Tribune (Chandigarh).
16. *Report prepared in Times New Roman, 12 pt font, with 1.5 line spacing.*